

Bio-Inspired Algorithms: An Introduction

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Course Summary

- ▶ Lecture 1 – Introduction, Evolutionary Computing & Genetic Algorithms
- ▶ Lecture 2 – Collective Intelligence
- ▶ Lecture 3 – Cellular Automata
- ▶ Lecture 4 – Artificial Immune Systems
- ▶ Lecture 5 – Neural Computation

Lecture 1 – Introduction, Evolutionary Computing & Genetic Algorithms

What is Bio-Inspired Computing?

Bio-inspired computing (a sub-field of *natural computing*) is the design of computational models, algorithms and architectures whose structure or behavior is abstracted from biological systems and natural processes. We have three complementary study areas [13, 16]:

- ▶ **Nature as inspiration:** bio-inspired mechanisms/algorithms (evolution, swarming, immunity) to solve real-world problems.
- ▶ **Nature as substrate:** compute *with* natural media (DNA [7], molecules, membranes [11]).
- ▶ **Nature as computation:** model and understand biological systems as information processing systems [5].

Here we will focus on **nature as inspiration**. The main idea is that computational algorithms can emerge from *simple local rules* and *populations of interacting agents*.

Why Look to Nature?

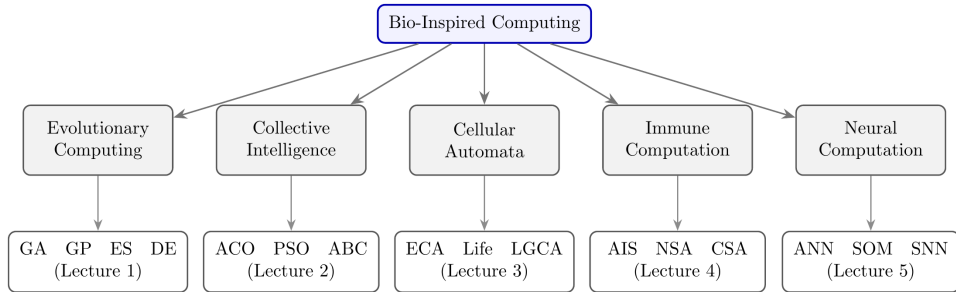
Natural systems routinely solve problems that are hard for classical algorithms:

- ▶ High-dimensional, non-linear, multimodal landscapes;
- ▶ Noisy, dynamic and uncertain environments;
- ▶ No gradient, no convexity, no closed form;
- ▶ Objective functions available only as a simulation or a physical experiment (*black-box optimisation*).

Desirable emergent properties [10, 15]:

- ▶ **Self-organisation** and adaptation;
- ▶ **Robustness** to failures;
- ▶ **Scalability** and parallelism;
- ▶ **Emergence** from local interactions.

Bio-Inspired Algorithms – Map of the Course



GA: Genetic Algorithms; GP: Genetic Programming; ES: Evolution Strategies; DE: Differential Evolution; ACO: Anti Colony Optimization; PSO: Particle Swarm Optimization; ABC: Artificial Bee Colony; ECA: Elementary Cellular Automata; LGCA: Lattice-Gas CA; AIS: Artificial Immune Systems; ANN: Artificial Neural Network; SOM: Self-Organising Map; SNN: Spiking Neural Network.

The Optimization Problem

Let S be a **search space** (binary strings, trees, graphs, etc.), $f : S \rightarrow \mathbb{R}$ be the **objective** function, and Ω be the **feasible region**. Every bio-inspired algorithm in this course is, formally, a stochastic process on S whose stationary behavior we would like to concentrate on $\arg \min f$.

$$\text{minimize}_{\mathbf{x} \in \Omega} f(\mathbf{x}), \quad \Omega = \{\mathbf{x} \in S : g_i(\mathbf{x}) \leq 0, h_j(\mathbf{x}) = 0\}.$$

What makes a problem hard: combinatorial explosion: $|S| = 2^n, n!$, etc.; multimodality: many local optima; noise: f observed as $f + \varepsilon$; dynamics: $f = f(\mathbf{x}, t)$.

What a metaheuristic assumes: f can be *evaluated*, not differentiated; a *neighborhood* or *variation operator* exists on S ; near-optimal is good enough.

Fitness Landscapes; Exploration vs Exploitation

A **fitness landscape** is the triple $(S, \mathcal{N}, f)$ where S is a search space, $\mathcal{N}$ is a neighborhood structure, and f is an objective function [12, 14]. The *same* f on a *different* neighborhood is a different landscape (this is why representation matters more than the choice of algorithm).

Exploration (diversification): Sample unvisited regions; protects against premature convergence, wastes evaluations.

Exploitation (intensification): Refine the best-so-far; converges fast, risks being trapped.

Examples: Genetic algorithm (exploration by mutation, large population; exploitation by selection pressure, crossover); Ant colony (exploration by pheromone evaporation; exploitation by pheromone reinforcement).

Evolutionary Computing

A **population** of candidate solutions evolves under *selection*, *variation* and *inheritance* [3, 12].

Generic evolutionary loop:

1. Initialize population randomly;
2. Evaluate fitness of each individual;
3. Select parents;
4. Recombine (crossover) and mutate;
5. Replace / form new generation;
6. Repeat until stopping criterion.

Mathematical modeling: Population: *multiset of points in S* ; Fitness: *objective function f* ; Natural selection: *biased sampling*; Recombination: *crossover*; Mutation: *random local perturbation*; Generations: *iteration*.

The Generic Evolutionary Algorithm (EA)

The generic evolutionary algorithm is the basis for several bio-inspired algorithms (from genetic to swarm algorithms), and they are particular instantiations of the following five choices:

The five design decisions: 1. Representation (how a solution is encoded); 2. Fitness (how quality is measured); 3. Variation (crossover and mutation); 4. Selection (who reproduces, who survives); 5. Population model (size, structure, replacement).

Algorithm 1: Generic evolutionary algorithm

Data: fitness f , population size μ , operators

Result: best individual found

```
1  $t \leftarrow 0$ ; initialise  $P_0$  at random
2 evaluate  $f$  on  $P_0$ 
3 while stopping criterion not met do
4    $Q \leftarrow \text{PARENTSELECTION}(P_t)$ 
5    $Q \leftarrow \text{RECOMBINE}(Q)$  with probability  $p_c$ 
6    $Q \leftarrow \text{MUTATE}(Q)$  with probability  $p_m$ 
7   evaluate  $f$  on  $Q$ 
8    $P_{t+1} \leftarrow \text{SURVIVORSELECTION}(P_t, Q)$ 
9    $t \leftarrow t + 1$ 
10 return  $\arg \max_{x \in \bigcup_t P_t} f(x)$ 
```

Genetic Algorithms (GA)

Introduced by **Holland** [3] and popularised by **Goldberg** [4].

- ▶ **Representation:** Fixed-length binary strings (*chromosomes*), $\mathbf{x} \in \{0, 1\}^n$.
- ▶ **Parent selection:** Fitness proportionate (roulette wheel selection).
- ▶ **Crossover:** One-point, applied with probability $p_c \approx 0.6\text{--}0.9$.
- ▶ **Mutation:** Independent bit flips with $p_m \approx 1/n$.
- ▶ **Survivor selection:** Generational replacement, often with elitism.

Selection I – Fitness Proportionate Selection

Roulette wheel: Individual i is chosen with probability

$$p_i = \frac{f_i}{\sum_{j=1}^{\mu} f_j}.$$

However, we have the following problems [4, 12]:

- ▶ **Premature convergence:** One super-individual with $f_i \gg \bar{f}$ takes over the population in a few generations.
- ▶ **Stagnation:** When f values become similar, $p_i \rightarrow 1/\mu$ and selection becomes a random walk.
- ▶ **Not invariant under $f \mapsto f + c$:** The algorithm's behavior changes if you add a constant to the objective.

Selection II – Ranking, Truncation, Tournament

- **Linear ranking:** Sort the population, then

$$p_i = \frac{1}{\mu} \left(s - (2s - 2) \frac{i - 1}{\mu - 1} \right), \quad 1 < s \leq 2,$$

where $i = 1$ is the best. Depends only on order, not on fitness magnitude.

- **Exponential ranking:** $p_i \propto c^{i-1}$, $0 < c < 1$.
- **Truncation** (μ, λ): Keep the best μ of λ offspring; standard in evolution strategies.
- **k -tournament:** Draw k individuals uniformly at random, keep the best.
 - ▶ selection pressure tuned by k ($k = 1$: random; $k = \mu$: elitist);
 - ▶ no global sorting, no global sums $\Rightarrow$ trivially parallel;
 - ▶ invariant under monotone transformations of f ;
 - ▶ probability that the best individual is chosen: $1 - (1 - 1/\mu)^k$.

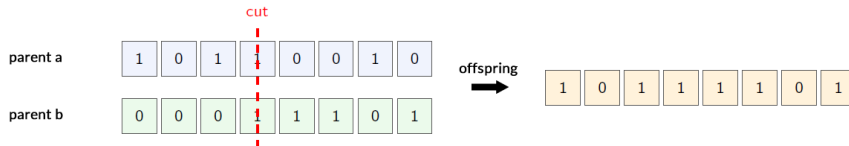
Variation I – Crossover (Binary and Integer Strings)

The purpose of crossover has been explained by two competing accounts: the *building-block hypothesis*, which posits that crossover assembles good partial solutions, and the *macromutation* view, which characterizes crossover as a large, population-directed jump [12].

Crossover methods for **binary and integer strings**:

- ▶ **One-point**: Cut at $c \sim \mathcal{U}\{1, n - 1\}$; strong *positional bias* (nearby genes stay together).
- ▶ **k-point**: k cuts; less positional bias.
- ▶ **Uniform**: Each gene taken from either parent with probability $\frac{1}{2}$; maximal mixing, disrupts long schemata.

Example (one-point crossover):



Variation I – Crossover (Real vectors and Permutations)

Crossover methods for **real vectors** and **permutations**:

Real vectors:

- ▶ **Arithmetic:** $\mathbf{x}' = \alpha \mathbf{x}_a + (1 - \alpha) \mathbf{x}_b$ (contracting, reduces variance).
- ▶ **Blend BLX- α :** sample uniformly in an interval extended by α beyond the parents (non-contracting).

Permutations:

- ▶ **PMX** (partially mapped), **OX** (order), **CX** (cycle): All designed so the offspring remains a valid permutation.
- ▶ **Edge recombination:** Preserves adjacency, which is what matters for the TSP.

Variation II – Mutation

Mutation guarantees **ergodicity**: with $p_m > 0$ every point of S is reachable, which is what makes asymptotic convergence proofs possible. Mutation methods:

- ▶ **Bit flip mutation** (rate p_m): Expected flips per string = np_m ; the classical choice $p_m = 1/n$ flips one bit on average.
- ▶ **Random reset**: One or more positions are randomly selected and assigned new values from the allowed domain, such as the positive integers.
- ▶ **Gaussian**: $x'_i = x_i + \sigma \mathcal{N}(0, 1)$; σ is the single most important parameter of a real-coded evolutionary algorithm.
- ▶ **Inversion mutation**: A substring of the chromosome is selected, reversed, and replaced.
- ▶ **Swap mutation**: Two genes are randomly selected and their values are exchanged.

Furthermore, the mutation rate plays a critical role in evolutionary search. If it is **too small**, the population may drift and eventually stall; if it is **too large**, the algorithm can degenerate into random search, reflecting the *error threshold* known from quasi-species theory. For a string of length n , the probability that a given individual remains unchanged is $(1 - p_m)^n \approx e^{-np_m}$; thus, with $p_m = 1/n$, this probability is approximately 0.37.

Population Models and Elitism

Replacement schemes

- ▶ *Generational*: $\lambda = \mu$ offspring replace the whole population.
- ▶ *Steady-state*: one or two offspring per iteration replace the worst (*generation gap* $G = \lambda/\mu$ small).
- ▶ $(\mu + \lambda)$: parents and offspring compete (elitist by construction).
- ▶ (μ, λ) , $\lambda > \mu$: parents die (allows temporary fitness decrease, better for self-adaptation).

Elitism

- ▶ Copy the best e individuals unchanged. Guarantees monotone best-so-far and is required by most convergence proofs.
- ▶ Too much elitism $\Rightarrow$ loss of diversity.

Structured populations

- ▶ *Island model*: subpopulations evolve separately and exchange migrants every τ generations.
- ▶ *Cellular EA*: individuals live on a grid and mate only with neighbors (a cellular automaton whose states are solutions (see Lecture 3)).

Example: 0/1 Knapsack Problem

Suppose we have $n = 5$ items with weights $\mathbf{w} = [2, 3, 4, 5, 1]$ and values $\mathbf{v} = [3, 4, 5, 6, 1]$, with capacity $W = 8$. Let's encode each chromosome $\mathbf{x} \in \{0, 1\}^5$, where $x_i = 1$ means item i is selected.

Fitness:

$$f(\mathbf{x}) = \sum_{i=1}^5 v_i x_i \quad \text{subject to} \quad \sum_{i=1}^5 w_i x_i \leq W, \quad f(\mathbf{x}) = 0 \text{ if constraint violated.}$$

Sample chromosomes:

Chromosome	Items selected	Total weight	Fitness
[1, 1, 0, 0, 1]	1, 2, 5	6	8
[1, 0, 1, 0, 0]	1, 3	6	8
[0, 1, 1, 0, 1]	2, 3, 5	8	10
[1, 1, 0, 1, 0]	1, 2, 4	10	0 (infeasible)

The chromosome $[0, 1, 1, 0, 1]$ achieves the best feasible fitness of 10.

The Schema Theorem

A **schema** H is a string over $\{0, 1, *\}$. It denotes the subset of the search space matching it, e.g. $H = 1**0$ matches $\{1000, 1010, 1100, 1110\}$. We denote by $o(H)$ the number of fixed positions (*order*) and by $\delta(H)$ the distance between the outermost fixed positions (*defining length*).

Theorem (Holland's schema theorem [3, 4]):

$$\mathbb{E}[m(H, t + 1)] \geq m(H, t) \frac{\hat{f}(H, t)}{\bar{f}(t)} \left[1 - p_c \frac{\delta(H)}{n - 1} - o(H) p_m \right],$$

where $m(H, t)$ is the number of representatives of H at generation t , $\hat{f}(H, t)$ its average fitness, and $\bar{f}(t)$ the population average.

It is an inequality on *one* generation, it ignores the gain from crossover, and it says nothing about convergence to the optimum.

Building Blocks, Linkage and Deception

Building-block hypothesis [4]: GAs work by discovering, preserving and recombining short, low-order, high-fitness schemata.

Linkage: Genes that must be co-adapted should be close together on the chromosome, otherwise crossover keeps breaking them apart.

Deceptive problems: A problem is *deceptive* when low-order schemata guide the search away from the global optimum. An example is the trap function:

$$f_{\text{trap}}(u) = \begin{cases} n & u = n \\ n - 1 - u & u < n \end{cases}$$

where u represents the number of ones. The optimum is $11 \dots 1$.

Deception, epistasis, and rugged landscapes constitute precisely the regime in which the No-Free-Lunch theorem applies [9].

Diversity: Preserving It, Measuring It

Premature convergence occurs when the population collapses toward a single basin of attraction before the budget is exhausted. Diversity is the resource that is being consumed.

Measures: Mean pairwise Hamming / Euclidean distance; population entropy per locus; fitness variance, number of distinct genotypes.

Restoring diversity: Increase p_m , restart, random immigrants; larger μ , weaker selection pressure.

Niching techniques:

- ▶ *Fitness sharing:* $f'_i = f_i / \sum_j \text{sh}(d_{ij})$ with $\text{sh}(d) = 1 - (d/\sigma_{\text{share}})^\alpha$ for $d < \sigma_{\text{share}}$.
- ▶ *Crowding / deterministic crowding:* offspring replace the most similar parent.
- ▶ *Clearing:* only the best few individuals per niche keep their fitness.
- ▶ *Island models:* geographic isolation.

The immune algorithms of Lecture 4 achieve niching *implicitly*, through affinity-proportional cloning.

Setting the Parameters

Parameter tuning (before the run): Factorial designs, racing, meta-evolutionary algorithms. The best values are problem- and budget-dependent.

Parameter control (during the run):

- ▶ *Deterministic*: $p_m(t) = p_0/\sqrt{t}$, cooling schedules.
- ▶ *Adaptive*: Feedback from the search (e.g. Rechenberg's 1/5 success rule [2]).
- ▶ *Self-adaptive*: Parameters encoded in the genome and evolved with it.

Rechenberg's 1/5 rule [2]: Measure the fraction p_s of successful mutations over a window:

$$\sigma \leftarrow \begin{cases} \sigma/c & p_s > 1/5 \quad (\text{expand}) \\ \sigma \cdot c & p_s < 1/5 \quad (\text{contract}) \\ \sigma & p_s = 1/5 \end{cases}$$

with $c \approx 0.817$. Derived for the sphere and corridor models.

Beyond GA: Evolution Strategies (ES)

Developed by Rechenberg and Schwefel for experimental optimization of physical designs [2].
The following notation is adopted:

- ▶ $(1 + 1)$: one parent, one offspring, elitist
- ▶ $(\mu + \lambda)$: elitist survivor selection
- ▶ (μ, λ) : non-elitist, $\lambda \geq 7\mu$ typical
- ▶ $(\mu/\rho^+; \lambda)$: ρ parents recombined

Self-adaptive mutation:

$$\begin{aligned}\sigma'_i &= \sigma_i \exp(\tau' \mathcal{N}(0, 1) + \tau \mathcal{N}_i(0, 1)) \\ x'_i &= x_i + \sigma'_i \mathcal{N}_i(0, 1)\end{aligned}$$

with $\tau \propto 1/\sqrt{2\sqrt{n}}$ and $\tau' \propto 1/\sqrt{2n}$.

(μ, λ) for self-adaptation: An individual carrying a bad σ must be allowed to die even if its objective value is momentarily good, otherwise poor strategy parameters survive on lucky fitness.

Beyond GA: Evolutionary Programming and Differential Evolution

Evolutionary programming (EP) [1]:

- ▶ Originally: Evolve finite-state machines for sequence prediction.
- ▶ No recombination: Mutation only; each parent produces exactly one offspring.
- ▶ Survivor selection by stochastic round-robin tournament.
- ▶ Modern EP is close to $(\mu + \mu)$ self-adaptive ES.

Differential evolution (DE) [8]:

For each target $\mathbf{x}_i$, choose distinct a, b, c :

$$\mathbf{v}_i = \mathbf{x}_a + F(\mathbf{x}_b - \mathbf{x}_c), \quad F \in (0, 2)$$

binomial crossover with rate CR , then *greedy* one-to-one replacement: $\mathbf{x}_i \leftarrow \mathbf{u}_i$ iff $f(\mathbf{u}_i) \leq f(\mathbf{x}_i)$.

Key insight: The population itself determines the mutation scale. Initially, difference vectors are long (exploration); as the population contracts, they decrease (exploitation).

Beyond GA: Genetic Programming (GP)

Genetic programming evolves computer programs, represented as syntax trees over a *function set* $\mathcal{F}$ and a *terminal set* $\mathcal{T}$ [6].

Requirements on $\mathcal{F}$ and $\mathcal{T}$:

- ▶ *Closure*: Every function must accept every possible argument value (hence *protected* division, protected log).
- ▶ *Sufficiency*: The target expression must be constructible.

Operators:

- ▶ *Crossover*: Exchange randomly chosen subtrees;
- ▶ *Mutation*: Replace a subtree by a new random one; point mutation of a node;
- ▶ *Initialisation*: *Grow*, *full*, or *ramped half-and-half*.

Convergence and the No-Free-Lunch Theorem

Convergence results:

- ▶ A GA with $p_m > 0$ and elitism is a Markov chain that visits the optimum with probability $\rightarrow 1$ as $t \rightarrow \infty$. Without elitism, it does not necessarily converge.
- ▶ Such results say nothing about *speed* (exhaustive search converges too).
- ▶ Useful runtime results exist only for simple functions (OneMax, LeadingOnes) and simple algorithms (1 + 1) EA.

Theorem (No Free Lunch [9]): Averaged over *all* objective functions $f : S \rightarrow Y$ with S, Y finite, every algorithm that never re-evaluates a point has the same expected performance:

$$\sum_f P(\mathbf{y} \mid f, m, a_1) = \sum_f P(\mathbf{y} \mid f, m, a_2).$$

The theorem doesn't say that all algorithms are equally good on the problems we care about; real problems are a vanishingly small, highly structured subset. It *does* say that performance claims must be tied to a problem class, and that any bias built into an operator is a hypothesis about that class.

Python – Example 1: A Genetic Algorithm for the 0/1 Knapsack

```
1 import numpy as np
2 rng = np.random.default_rng(0)
3
4 n, C = 40, 500                                # items, knapsack capacity
5 w = rng.integers(10, 60, n)                   # weights
6 v = rng.integers(10, 90, n)                   # values
7
8 def fitness(P):                                # P: (mu, n) binary population
9     load, val = P @ w, P @ v
10    return np.where(load ≤ C, val, val - 5.0 * (load - C)) # penalty
11
12 def tournament(P, f, k=3):
13     idx = rng.integers(0, len(P), (len(P), k))
14     return P[idx[np.arange(len(P)), np.argmax(f[idx], axis=1)]]
15
16 def uniform_crossover(P, pc=0.9):
17     Q = P.copy()
18     for i in range(0, len(P) - 1, 2):
19         if rng.random() < pc:
20             m = rng.random(P.shape[1]) < 0.5
21             Q[i], Q[i + 1] = np.where(m, P[i], P[i + 1]), np.where(m, P[i + 1], P[i])
22     return Q
23
24 def mutate(P, pm=None):
25     pm = 1.0 / P.shape[1] if pm is None else pm
26     return np.where(rng.random(P.shape) < pm, 1 - P, P)
```

Python – Example 1: A Genetic Algorithm for the 0/1 Knapsack (cont.)

The Evolutionary Loop:

```
28 mu, gens = 100, 200
29 P = rng.integers(0, 2, (mu, n))
30 best = None
31 for g in range(gens):
32     f = fitness(P)
33     elite = P[np.argmax(f)].copy()           # elitism: keep the champion
34     if best is None or f.max() > fitness(best[None])[0]:
35         best = elite
36     P = mutate(uniform_crossover(tournament(P, f)))
37     P[0] = elite
38     print("best value =", best @ v, " weight =", best @ w, " / C =", C)
```

Output: best value = 1281, weight = 499, C = 500. A greedy value/weight heuristic gives a strictly worse packing on this instance. Note the vectorized tournament: draw a $\mu \times k$ index matrix, then take the row-wise $\arg\max$ of the fitness.

Python – Example 2: Differential Evolution on Rastrigin Function

```
1 import numpy as np
2 rng = np.random.default_rng(1)
3
4 def rastrigin(X):
5     # global minimum f(0)=0, many local minima
6     return 10 * X.shape[1] + np.sum(X**2 - 10 * np.cos(2 * np.pi * X), axis=1)
7
8 def de(f, d=10, mu=60, F=0.5, CR=0.9, gens=1500, lo=-5.12, hi=5.12):
9     P = rng.uniform(lo, hi, (mu, d)) # initial population
10    fit = f(P)
11    for g in range(gens):
12        for i in range(mu):
13            a, b, c = rng.choice(np.delete(np.arange(mu), i), 3, replace=False)
14            V = P[a] + F * (P[b] - P[c]) # DE/rand/1 mutation
15            V = np.clip(V, lo, hi)
16            j = rng.integers(d) # ensure ≥ 1 inherited gene
17            m = (rng.random(d) < CR); m[j] = True
18            U = np.where(m, V, P[i]) # binomial crossover
19            fu = f(U[None])[0]
20            if fu ≤ fit[i]: # greedy one-to-one selection
21                P[i], fit[i] = U, fu
22    k = np.argmin(fit)
23    return P[k], fit[k]
24
25 x, fx = de(rastrigin)
26 print("f* = %.6f    ||x*|| = %.3e" % (fx, np.linalg.norm(x)))
```

Output: $f^* = 0.000007$, $\|x^*\| = 1.8 \times 10^{-4}$. In $d = 10$ the Rastrigin function has of the order of 10^{10} local minima on $[-5.12, 5.12]^{10}$.

Homework (Lecture 1)

1. On the genetic algorithm:

- ▶ Replace the penalty by a *repair* operator that drops the item with the worst value/weight ratio until the load is feasible. Which converges faster?
- ▶ Set $k = 1$ (no selection pressure), then $k = \mu$ (extreme pressure). Observe stagnation and premature convergence respectively.
- ▶ Plot the mean pairwise Hamming distance per generation (diversity curve).

2. On the differential evolution algorithm:

- ▶ Sweep $F \in [0.3, 1.0]$ and $CR \in \{0.1, 0.5, 0.9\}$ on Rastrigin and on the sphere. Separable problems prefer small CR .
- ▶ Implement DE/best/1/bin by replacing $\mathbf{x}_a$ with the current best. Faster, but check how often it now gets trapped.
- ▶ Reduce μ to 10: the difference vectors lose diversity and the search stalls (the population is the step-size distribution).

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